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IIT ACADEMY
INDIA

40 YEARS OF EXCELLENCE

Section: Senior **TOPIC: Qualitative analysis** **Date: 04-07-2020**

SINGLE ANSWER TYPE

1. SO_2 and CO_2 both turn lime water (X) milky, SO_2 also turns $K_2Cr_2O_7/H^+$ (Y) green while O_2 is soluble in pyrogallol (Z) turning it black. These gases are to be detected in order by using these reagents. The order is:
(A) (X), (Y), (Z) (B) (Y), (X), (Z)
(C) (X), (Z), (Y) (D) The correct order cannot be predicted.

2. Colourless salt (A) + dil. H_2SO_4 or CH_3COOH + KI $\longrightarrow$ blue colour with starch. (A) can be
(A) K_2SO_3 (B) Na_2CO_3 (C) NH_4NO_2 (D) NH_4Cl

3. Zinc. pieces are added to acidified solution of SO_3^{2-}. Gas liberated can :
(A) turn lead acetate paper black (B) turn lime water milky
(C) give both of the above tests (D) give none of the above tests

4. A mixture when rubbed with dilute acid smells like vinegar. It contains :
(A) sulphite (B) nitrate (C) nitrite (D) acetate

5. When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with conc. H_2SO_4, deep red vapours are obtained. This is due to the formation of :
(A) Chromous chloride (B) Chromyl chloride (C) Chromic chloride (D) Chromic sulphate

6. AgCl dissolves in ammonia solution giving :
(A) Ag^+, NH_4^+ and Cl^- (B) $Ag(NH_3)^+$ and Cl^- (C) $Ag_2(NH_3)^{2+}$ and Cl^- (D) $Ag(NH_3)_2^+$ and Cl^-

7. A colourless solution of a compound gives a precipitate with $AgNO_3$ solution but no precipitate with a solution of Na_2CO_3. The action of concentrated H_2SO_4 on the compound liberates a suffocating reddish brown gas. The compound is :
(A) $Ba(NO_3)_2$ (B) $CaCl_2$ (C) NaI (D) NaBr

8. When Cl_2 water in excess is added to a salt solution containing chloroform, chloroform layer turns pale yellow. Salt contains :
(A) Br^- (B) I^- (C) NO_3^- (D) S^{2-}

9. An aqueous solution of salt containing an acidic radical X^- reacts with sodium hypochlorite in neutral medium. The gas evolved produces blue black colour spot on the starch paper. The anion X^- is :
(A) CH_3COO^- (B) Br^- (C) I^- (D) NO_2^-

10. Nitrate is confirmed by ring test. The brown colour of the ring is due to formation of :
(A) Ferrous nitrite (B) Nitroso ferrous sulphate
(C) Ferrous nitrate (D) $FeSO_4.NO_2$.

11. A metal X on heating in nitrogen gas gives Y.Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a intense blue colour. Y is :
(A) $Mg(NO_3)_2$ (B) Mg_3N_2 (C) NH_3 (D) MgO

12. A metal chloride solution on mixing with K_2CrO_4 solution gives a yellow precipitate soluble in aqueous sodium hydroxide. The metal may be
(A) Mercury (B) Zinc (C) Silver (D) Lead

13. In borax bead test which compound is formed ?
(A) Orthoborate (B) Metaborate (C) Double oxide (D) Tetraborate

14. Aqueous Solution of $BaBr_2$ gives yellow precipitate with :
(A) K_2CrO_4 (B) $AgNO_3$ (C) $(CH_3COO)_2Pb$ (D) (A) and (B) both

15. The addition of K_2CO_3 (aq) to the following solution is expected to produce a precipitate in every case but that one which does not produce precipitate is :

(A) $BaCl_2$(aq) (B) $CaBr_2$(aq) (C) Na_2SO_4(aq) (D) $Pb(NO_3)_2$ (aq)

16. If crimson flame is given when an inorganic mixture is tested by flame test, it may be due to the presence of

(A) Chromium (B) Strontium (C) Barium (D) Calcium

KEY

1.	B	2.	C	3.	A	4.	D	5.	B	6.	D	7.	D
8.	A	9.	C	10.	B	11.	B	12.	D	13.	B	14.	D
15.	C	16.	B										